

Sherif Abd El-Rady

AI Engineer · Computer Vision & Deep Learning

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SUMMARY

AI Engineer specializing in end-to-end computer vision and deep learning systems — from model architecture and training to ONNX optimization and production deployment. Delivered 14 complete CV systems spanning medical imaging, real-time detection, OCR, multimodal search, and autonomous perception. Available for remote freelance and contract work.

TECHNICAL SKILLS

Core ML / CV: PyTorch · TorchVision · ONNX Runtime · OpenCV · NumPy · scikit-learn

CV Domains: Object Detection · Segmentation · Face Recognition · OCR · Depth Estimation · Pose

Serving: FastAPI · Triton Inference Server · Docker · REST APIs · INT8 Quantization

Languages: Python (expert) · C++ (OpenCV) · SQL · Bash

Infra / MLOps: Git · GitHub Actions · Docker Compose · Linux · CUDA

Frameworks: YOLOv8 · DenseNet · ResNet · ViT · CLIP · CRNN · U-Net · ArcFace · SlowFast

SELECTED PROJECTS

MediScan — Medical Image Classification | PyTorch · DenseNet-121 · Focal Loss · Grad-CAM

97.3% accuracy · AUC 0.99 · F1 0.96 · NIH Chest X-Ray 14 (112K images)

Multi-label chest X-ray classifier. Focal Loss ($\alpha=0.25$, $\gamma=2.0$) over BCE lifted COVID-19 sensitivity from 78.3% to 91.4%. Grad-CAM overlays for clinical explainability. Deployed as REST API.

InferEdge — Production ML Inference Server | FastAPI · ONNX Runtime · Triton · Docker

<8ms P99 latency · 500+ req/min · 3.1× speedup via INT8 quantization

Production inference pipeline for YOLOv8-Large. Dynamic batching via Triton reduced P99 by 40% versus naive FastAPI. Memory pre-allocation eliminated cold-start spikes.

UrbanSense — Real-Time Object Detection | YOLOv8 · ONNX · INT8 · FastAPI · OpenCV

mAP 0.891 · 45 FPS @ 1080p · 8ms inference

Custom YOLOv8 for urban scene understanding. Modified anchor-free head (P3 stride 8→6) for small-object recall. INT8 ONNX export: 3.1× speedup with only 0.6% mAP drop.

DocuAI — OCR & Document Intelligence | CRNN · CRAFT · CTC · PyTorch

96.4% word accuracy · Arabic + Latin · end-to-end pipeline

CRNN + CRAFT text detection for mixed-language documents. Custom right-to-left augmentation for Arabic. ONNX-optimized for edge deployment.

VisionLang — Multimodal Visual Search | CLIP (ViT-L/14) · FAISS · FastAPI

92.4% Recall@5 · sub-100ms query latency

Text-to-image and image-to-image retrieval using OpenAI CLIP embeddings indexed in FAISS. Supports incremental index updates and reranking.

SegForge — Semantic Segmentation | U-Net · ASPP · PyTorch · Cityscapes

mIoU 0.843 · pixel accuracy 94.1% · 19 classes

Hybrid U-Net + ASPP architecture. ASPP (rates 6/12/18) adds multi-scale context; U-Net skip connections preserve spatial boundaries. Outperforms DeepLabV3+ by +3.1% mIoU.

Additional: FaceVault (ArcFace, TAR 99.2%) · TrackMaster (DeepSORT, MOTA 0.812) · ActionNet (SlowFast, Top-1 87.4%) · SynthVision (StyleGAN2, FID 18.4) · SuperRes (ESRGAN, PSNR 32.7 dB) · PoseForge (ViTPose, AP 77.1%) · DepthPro (DPT, δ_1 93.2%) · AutoPerception (BEV, mAP 0.624 nuScenes)

EXPERIENCE

Independent AI Engineer · Freelance / Remote

2023 – Present

End-to-end delivery of computer vision and deep learning systems for clients across HealthTech, document automation, and smart city domains. Full ownership from data pipeline through model training, optimization, and production deployment.

EDUCATION

Bachelor's Degree — Computer Science / Engineering

Egypt

Full project code · github.com/sherifabdelrady | Open to remote freelance and contract work